Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ____S0/0/0________

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? __129________

Does R2 show up as an OSPF neighbor on R1? __Yes______

Does R2 show up as an OSPF neighbor on R3? ___No_____

What does this information tell you?

R2 has a direct neighbor relationship only with R1, and no with R3. This is because the S0/0/0 interface of R2 connecting to R1 has executed "no passive-interface" operation, so this interface can send and receive OSPF messages normally. The S0/0/1 interface connecting to R3 remains in a passive state. Furthermore, the cumulative cost is 129, because traffic from R3 to 192.168.2.0/24 requires passing through two T1 serial links (64*2), plus R2 gigabit 0/0 LAN link (cost=1).

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

router ospf 1
no passive-interface s0/0/1

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ___S0/0/1_________

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

One T1 serial link (64) + R2 Gigabit 0/0 LAN link (1) = 65

Is R2 listed as an OSPF neighbor to R3? ___ Yes _____

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Because this is compatible with the high-speed link environment of modern networks. 2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

To 192.168.3.0/24 = 781 (R1 S0/0/1) +1 (R3 G0/0) = 782
To 192.168.23.0/30 =781 (R1 S0/0/1) + 64 (R3 S0/0/1) = 845

2

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

781 + 781 = 1562 As there are two serial links, each costing 781. 2

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Because the experiment above increased the cost of the S0/0/1 interface directly connecting R1 to R3 (R1-S0/0/1 + R3-G0/0 = 1566), the total cost of the direct path via R3 was higher than the indirect path via R2 (R1 + R2 + R3-G0/0 = 1,563). OSPF always selects the path with the minimum cumulative cost as the optimal route.

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   This is because it ensures the uniqueness of the router ID, preventing situations such as neighbor relationship establishment failures and abnormal routing information exchange. Also, manual configuration and router setup based on loopback interfaces are unaffected by interface status. This improves troubleshooting efficiency.

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2. Why is the DR/BDR election process not a concern in this lab?

   DR/BDR election is only required in multi-access network types. The above exercise only applies to point-to-point links.

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3. Why would you want to set an OSPF interface to passive?

   Setting an OSPF interface as a passive interface prevents that interface from sending and receiving OSPF control traffic. This reduces network-irrelevant traffic and lowers the load on the router. In addition, this improves network security by establishing adjacencies only on necessary interconnecting interfaces, limiting the propagation of control traffic.

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4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   OSPF is based on area as its primary structural design. Area 0 serves as the backbone area. Each area also independently maintains its LSDB to run the SPF algorithm. This reduces flooding and computational overhead, making it more suitable for scalable networks.

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5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

   network 10.0.0.0 0.255.255.255 area 0

   4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   EIGRP. Because routers do not compare metrics across different protocols, but rather by administrative distance (AD). Here, EIGRP has the lowest AD, so it is chosen.

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